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### Determining traffic violation hotspots

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#### Abstract

System and methods for determining traffic violation hotspots based on roadway feature and/or sensor data. Traffic violation hotspots include, for example areas where traffic violations are more likely to occur such as traffic light jumping, wrong way driving, over speeding, not wearing seatbelt, avoiding stop signs and red lights, distracted driving, passing other vehicles in a no-passing zone, among others. Embodiments further provide predictions for traffic accidents hotspots based on the presence or absence of the traffic violation hotspots.

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## Background/Summary

### FIELD

(1) The following disclosure relates to navigation devices or services.

### BACKGROUND

(2) During operation of a vehicle, there are driving circumstances, referred to as risk situations that increase the likelihood of accidents. A risk situation may be, for example, any circumstance leading to any type of contact with moving or fixed objects that causes damage. In certain areas, risky traffic situations may be more or less prevalent; for example, less prevalent driving close to schools or in residential areas, but more prevalent driving through intersections and roundabouts, etc. To reduce the burden of traffic crashes, it is critical to examine the time, locations, and circumstances of crashes where they occur more frequently. Locations that have clusters of high concentrations of crashes are commonly known as crash-prone locations or hotspots. Hotspot identification is a vital task for road traffic safety programs.

(3) One key aspect is identifying and understanding which driver behaviors contribute to traffic crashes and which road and built environment factors prompt these behaviors to emerge. Traffic violations, for example caused by aggressive driving behavior, are often seen as a primary contributor to traffic crashes. Violations are either caused by an unintentional or deliberate act of drivers that jeopardize the lives of fellow drivers, pedestrians, and property. The identification and understanding of where traffic violations occur may help engineers and policy makers design safer roads and policies to improve traffic safety.

### SUMMARY

(4) In an embodiment, a method is provided for determining traffic violation hotspots. The method includes acquiring data related to a location of a roadway, generating, using at least one model trained using machine learning, a probability score on how probable the location is to be a traffic violation hotspot based on the acquired data, determining, that the probability score exceeds a threshold score, acquiring feature data about the location and a region encompassing the location, generating, by a machine learning model, a probability of an accident occurring at the location, and generating an alert for the location based on the probability.

(5) In an embodiment, a system is provided for determining traffic violation hotspots. The system includes one or more navigation devices and a mapping system. The one or more navigation devices are configured to acquire sensor data related a plurality of locations on a roadway. The mapping system includes a geographic database configured to store historical mapping data for the plurality of locations. The mapping system further includes a server configured to configure and store a first machine learning model configured to generate a first probability score based on the historical mapping data and to configure and store a second machine learning model configured to generate a second probability score based on the sensor data. The server is configured to provide the first probability score, the second probability score, or the first probability score and the second probability score to the one or more navigation devices.

(6) In an embodiment, an apparatus for determining traffic violation hotspots is provided. The apparatus includes at least one processor; and at least one memory including computer program code for one or more programs; the at least one memory configured to store the computer program code configured to, with the at least one processor, cause the at least one processor to: acquire data related to a location of a roadway; generate, using at least one model trained using machine learning, a probability score on how probable is the location is likely to be a traffic violation

hotspot based on the acquired data; determine, that the probability score exceeds a threshold score; acquire feature data about the location and a region encompassing the location; generate, by a machine learning model, a probability of an accident happening at the location; and generate an alert for the location.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the present invention are described herein with reference to the following drawings.

(2) FIG. 1 depicts an example system for determining traffic violation hotspots according to an embodiment.

(3) FIG. 2 depicts an example depiction of a map with a traffic violation hotspot according to an embodiment.

(4) FIG. 3 depicts an example device of FIG. 1 according to an embodiment.

(5) FIG. 4 depicts an example workflow for determining traffic violation hotspots according to an embodiment.

(6) FIG. 5 depicts an example region of a geographic database.

(7) FIG. 6 depicts an example geographic database of FIG. 1.

(8) FIG. 7 depicts an example structure of the geographic database.

(9) FIG. 8 depicts an example autonomous vehicle according to an embodiment.

### DETAILED DESCRIPTION

(10) Embodiments provide systems and methods for determining traffic violation hotspots based on roadway feature and/or sensor data. Traffic violation hotspots include, for example areas where traffic violations are most likely to occur such as traffic light jumping, wrong way driving, over speeding, not wearing seatbelt, avoiding stop signs and red light, distracted driving, passing other vehicles in a no-passing zone, among others. Embodiments further provide predictions for traffic accidents hotspots based on the presence or absence of the traffic violation hotspots.

(11) In an embodiment, a vehicle's sensor data is acquired, and the vehicle's location is map matched. Traffic violations are quantified for a given time and area. Traffic violation statistics (mean, median, standard deviations) are calculated for various times (e.g., each hour) for each given area. If the statistics of the traffic violation for the given area and for the given time is same (or less) as that of the nation (or county or state) statistics, then the area is not flagged. If the statistics of the traffic violation for the given area and for the given time is greater than that of the nation (or county or state) statistics, then the region (or area) is flagged as mild traffic violation zone, moderate traffic violation zone, or high traffic violation zone. The mild, moderate and high traffic violation zone may be represented by different colors on the map. When the vehicle is approaching the traffic violation hotspot, the system may provide an alert to the vehicle or the operator of the vehicle. Embodiments allows users to benefit from increased safety by dynamically determining the areas where traffic violations are most likely to occur.

(12) Traffic accidents and traffic violations may be directly connected. Risky and aggressive driving behavior of drivers is regarded as the one of the leading causes of road traffic accidents. In some estimates, driver related factors account for more than 90% of crash occurrences. Some of the prevailing driver factors in this regard include traffic violations such as: distracted driving (for example, mobile phone use), drunk driving, driving under fatigue, risky and aggressive driving attitudes that leads to various traffic violations i.e., over-speeding, red-light crossing, non-compliance with pedestrian signals, road markings, etc. These traffic violations and others are reported to have a strong bearing on crash occurrences as well as associated crash severities.

(13) As an example, over-speeding is one of the most prevailing traffic violations encountered

which has resulted in a large number of severe and fatal crashes. Speeding not only increases the risk of fatal traffic accidents, but also makes them worse. Studies also suggest that drivers are much more likely to crash while talking on the phone. Another form of distraction which causes many accidents is “rubbernecking,” or turning to look at another accident while passing by. Any time a motorist is not focused solely on the road and other vehicles, they are driving while distracted and the consequences are dangerous.

(14) Accidents may also cause traffic violations which in return may cause more accidents. In an example, if an accident is detected in a given area on a given lane, it increases the risk of people using the opposite lane to go over it (depending on the area, validated by probe data in that area). Based on this information, proposed embodiments predict that when an accident happens in the future on that same lane and conditions are similar (traffic, weather, time of day, etc), then there will likely be people attempting to overtake on the opposite lane, leading to increased safety risks. If there is congestion on one side of the road and the other side of the road has a sparse vehicle then the motorist may have the tendency to drive from the wrong side of the road and beat the congestion. Similarly, in a commercial area (where there are offices) there may be higher cases of traffic light jumping during peak hours (probable reason—getting late for work). Such patterns can be identified from this traffic violation hotspot. Identification of predictors can help in reducing traffic related risks.

(15) The following embodiments relate to several technological fields including but not limited to navigation, autonomous driving, assisted driving, traffic applications, and other location-based systems. In each of the technologies of navigation services, autonomous driving, assisted driving, traffic applications, and other location-based systems, improved identification and generation of three-dimensional building structures improves the ability of the mapping system to provide a safe and satisfactory trip. In addition, users of navigation, autonomous driving, assisted driving, traffic applications, and other location-based systems are more willing to adopt these systems given the technological advances in improved safety, visualization, and understanding of the roadway.

(16) FIG. 1 depicts a system for determining traffic violation and traffic accident hotspots. The system includes at least one or more devices **122**, a network **127**, and a mapping system **121**. The mapping system **121** may include a database **123** (also referred to as a geographic database **123** or map database) and a server **125**. Additional, different, or fewer components may be included. The mapping system **121** is configured to generate prediction models for traffic violation hotspots using machine learning techniques and historical data collected by the one or more devices **122** and data stored in the geographic database **123**. The mapping system **121** is further configured to generate prediction models for traffic accidents using machine learning techniques, the traffic violation hotspots predictions, and data stored in the geographic database **123**. A device **122** traveling the roadway applies the prediction models in order to understand the upcoming roadway based on sensor data and/or roadway feature data collected by the device **122**.

(17) The one or more devices **122** may include probe devices **122**, probe sensors, IoT (internet of things) devices **122**, or other devices **122** such as personal navigation devices **122** or connected vehicles. The device **122** may be a mobile device or a tracking device that provides samples of data for the location of a person or vehicle. The devices **122** may include mobile phones running specialized applications that collect location data as the devices **122** are carried by persons or things traveling a roadway system. The one or more devices **122** may include traditionally dumb or non-networked physical devices and everyday objects that have been embedded with one or more sensors or data collection applications and are configured to communicate over a network **127** such as the internet. The devices **122** may be configured as data sources that are configured to acquire image data and/or roadway feature data. These devices **122** may be remotely monitored and controlled. The devices **122** may be part of an environment in which each device **122** communicates with other related devices in the environment to automate tasks. The devices **122** may communicate sensor data to users, businesses, and, for example, the mapping system **121**.

(18) The one or more devices **122** are configured to collect data related to traffic violations by, for example, a vehicle that the device **122** is embedded with or otherwise traveling with, or for example, other vehicles on the roadway with sensor range of a device **122**. The traffic violations may include any potential traffic violation for a given area. Some examples include speeding, failure to stop at a red light, failure to signal, and reckless driving that includes tailgating, illegal passing, driving the wrong way down a one-way street, passing on a curve, using the opposing traffic lanes to pass drivers, weaving in and out of traffic, and driving on the shoulder among other maneuvers.

(19) Traffic violations may be self-identified or may be reported on by other vehicles or sensors. In an example, a device **122** may detect that the device **122** or vehicle is exceeding a speed limit for certain area. The device **122** records the speed, the location, sensor data, and other feature data about the event. In another example, a device **122** may detect that another vehicle failed to stop at a red light. The device **122** records the location, sensor data (for example image data), and other feature data about the event. In both cases, the device **122** may package the event data into a report that details not only the traffic violation, but also sensor and roadway feature data that describes the circumstances (herein referred to traffic violation data). The device **122** is configured to transmit the traffic violation data to the server **125** or mapping system **121** for aggregation with other traffic violation data in order to train or configure a traffic violation prediction model.

(20) One or more of the devices **122** may also be configured to provide probe reports to the mapping system **121** while traversing a roadway network. The probe reports may be similar to the traffic violation data, for example, including sensor and feature data about the roadway. Probe reports, however, may not indicate a traffic violation. Each vehicle and/or mobile device **122** may include position circuitry such as one or more processors or circuits for generating probe data. The probe data may be generated by receiving Global Navigation Satellite System (GNSS) signals and comparing the GNSS signals to a clock to determine the absolute or relative position of the vehicle and/or mobile device **122**. The probe data may be generated using embedded sensors or other data relating to the environment of a vehicle or device **122**. The probe data may include a geographic location such as a longitude value and a latitude value. In addition, the probe data may include a height or altitude. The probe data may be collected over time and include timestamps. In some examples, the probe data is collected at a predetermined time interval (e.g., every second, every 100 milliseconds, or another interval). The probe data may also describe the speed, or velocity, of the mobile device **122**. The speed may be determined from the changes of position over a time span calculated from the difference in respective timestamps. The time span may be the predetermined time interval, that is, sequential probe data may be used. In some examples, the probe data is collected in response to movement by the device **122** (i.e., the probe report's location information when the device **122** moves a threshold distance). The predetermined time interval for generating the probe data may be specified by an application or by the user. The interval for providing the probe data from the mobile device **122** to the server **125** may be the same or different than the interval for collecting the probe data. The interval may be specified by an application or by the user.

(21) The one or more devices **122** may also be configured to acquire image data using one or more cameras embedded in or in communication with the one or more devices **122**. The image data may be included with the traffic violation data and may be transmitted to the mapping system **121** for storage in the geographic database **123** and processing by the server **125**. The image data may include metadata, for example, attributes about an image, such as its height and width, in pixels. The metadata may describe the content of the image, the date and time of the image, etc.

(22) The one or more devices **122** may be in communication with the sensors or may directly acquire information. In an example, the one or more devices **122** may communicate with a vehicle's engine control unit (ECU) that controls a series of actuators on an internal combustion engine to ensure optimal engine performance. The ECU data may be provided in the traffic violation reports.

In another example, the one or more devices **122** may communicate with a seat belt sensor that detects when a metal buckle of the seat belt is inserted into the female portion of the seat belt. If a detects an occupant seated in the driver or front passenger seat while the vehicle is in motion, but the seat belt is not buckled in, the device **122** may identify that a seat belt violation has occurred. A headlight sensor, wiper sensor, fog light sensor, etc. may also communicate with the one or more devices **122**. These sensors may provide data that details a violation or data that provides background data for the traffic violation event.

(23) The one or more devices **122** are also configured to identify and report traffic accidents. Similar to the traffic violations, traffic accident data may be reported by a vehicle involved in the accident or by another vehicle, device, or sensor. Traffic accident data may also be provided by a regulatory body. The traffic accident data may include numerous different data from different sensors such as the location, the ECU, image/video data, environmental data, etc.

(24) The one or more devices **122** may communicate the traffic violation data relating to traffic violations and the traffic accident data relating to traffic accidents to the server **125** or mapping system **121**. The information may be anonymous, for example detailing the violation, location, and other sensor data, but without personal or privacy related data. The traffic violation data is thus not used for citations or prosecutions, but rather as an important data point in understanding where, when, and how traffic accidents occur. The traffic violation data may also be combined with actual traffic violation citations recorded, for example, by an authority or roadway governance body.

(25) To communicate with the devices **122**, systems or services, the mapping system **121** is connected to the network **127**. The mapping system **121** may receive or transmit data through the network **127**. The mapping system **121** may also transmit paths, routes, or traffic accident/traffic violation data through the network **127**. The mapping system **121** may also be connected to an OEM cloud that may be used to provide mapping services to vehicles via the OEM cloud or directly by the mapping system **121** through the network **127**. The network **127** may include wired networks, wireless networks, or combinations thereof. The wireless network may be a cellular telephone network, LTE (Long-Term Evolution), 4G LTE, a wireless local area network, such as an 802.11, 802.16, 802.20, WiMAX (Worldwide Interoperability for Microwave Access) network, DSRC (otherwise known as WAVE, ITS-G5, or 802.11p and future generations thereof), a 5G wireless network, or wireless short-range network such as Zigbee, Bluetooth Low Energy, Z-Wave, RFID and NFC. Further, the network **127** may be a public network, such as the Internet, a private network, such as an intranet, or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to transmission control protocol/internet protocol (TCP/IP) based networking protocols. The devices **122** may use Vehicle-to-vehicle (V2V) communication to wirelessly exchange information about their speed, location, heading, and roadway conditions with other vehicles, devices **122**, or the mapping system **121**. The devices **122** may use V2V communication to broadcast and receive omni-directional messages creating a 360-degree “awareness” of other vehicles in proximity of the vehicle. Vehicles equipped with appropriate software may use the messages from surrounding vehicles to determine potential threats or obstacles as the threats develop. The devices **122** may use a V2V communication system such as a Vehicular ad-hoc Network (VANET).

(26) The geographic database **123** is configured to store and provide information to and from at least the mapping system **121**, server **125**, and devices **122**. The geographic database **123** may store and organize the traffic accident/traffic violation data received from the devices **122** and/or processed by one or more models provided by the mapping system **121**. The geographic database **123** may include one or more indexes of geographic data. The indexes may include various types of indexes that relate the different types of data to each other or that relate to other aspects of the data contained in the geographic database **123**. The indexes may include, for example, data relating to points of interest or roadway features. The point of interest data may include point of interest records including, for example, a type (e.g., the type of point of interest, such as restaurant, fuel

station, hotel, city hall, police station, historical marker, ATM, golf course, truck stop, vehicle chain-up stations etc.), location of the point of interest, a phone number, hours of operation, etc. The geographic database **123** provides data for the traffic violation and accident models. The geographic database **123** may be maintained by a content provider (e.g., a map developer). By way of example, the map developer may collect geographic data to generate and enhance the geographic database **123**. The map developer may obtain data from sources, such as businesses, municipalities, or respective geographic authorities. In addition, the map developer may employ field personnel to travel throughout the geographic region to observe features and/or record information about the roadway.

(27) The mapping system **121** may include multiple servers **125**, workstations, databases, and other machines connected together and maintained by a map developer. The mapping system **121** may be configured to acquire and process data relating to roadway or vehicle conditions. For example, the mapping system **121** may receive and input data such as vehicle data, user data, weather data, road condition data, road works data, traffic feeds, etc. The data may be historical, real-time, or predictive. The data may be stored in an HD map, in a location graph, or in the geographic database **123** for use in location-based services and navigation-based services. The mapping service may also provide information generated from attribute data included in the database **123**.

(28) The server(s) **125** may be a host for a website or web service such as a mapping service and/or a navigation service. The mapping service may provide standard maps or HD maps generated from the geographic data of the database **123**, and the navigation service may generate routing or other directions from the geographic data of the database **123**. The mapping service may also provide information generated from attribute data included in the database **123**. The server **125** may also provide historical, future, recent or current traffic conditions for the links, segments, paths, or routes using historical, recent, or real-time collected data. The server **125** is configured to communicate with the devices **122** through the network **127**. The server **125** is configured to receive a request from a device **122** for a route or maneuver instructions and generate one or more potential routes or instructions using data stored in the geographic database **123**. The server **125** may also be configured to provide up to date information and maps to external geographic databases or mapping applications.

(29) In an embodiment, the server **125** is configured to receive traffic violation data from devices **122**. The server **125** is configured to store, adjust, and implement at least one machine learning model that is configured to dynamically determine the areas where traffic violations are most likely to occur. In an embodiment, a model is used that is configured to predict traffic violation for a particular region based on only map features. The map features of interest may include a functional class of a road, category of a road (traffic light, roundabout, other junctions, toll plaza), a number of lanes, a speed limit, a travel direction, a presence of a physical divider (yes/no), among other roadway features. Using these features, a machine learning model is trained that predicts traffic violation hotspots given new feature data. Once the model is trained, it is deployed on the mapping system **121**. When a vehicle approaches a particular place of interest (intersection, roundabout, junctions), the map features of those locations are extracted and sent as an input to the trained machine learning model that is deployed on the cloud. The machine learning model provides a probability score (in a range of 0 to 1) on how probable is the region likely to be a traffic violation hotspot. If the probability is greater than, for example, 75% then the vehicle or the driver is warned about it. Based on the training data that included annotated data such as known traffic violation data, the model learns which map features (among the ones listed above but also any other map feature data available) are the most relevant based on the correlations between those features and specific or general traffic violations. Once those map features are identified and properly weighted, the model applies the correlations to make the predictions.

(30) In an embodiment a model is used that is configured to predict traffic violation for a particular region based on only sensor data. The vehicle sensor features are collected for a radius, for example



10, 50, 100, 500 meters, from the location of the vehicle. In an example, if a location (x,y) is to be evaluated whether it is a traffic hot-spot or not then all the vehicle sensor data is collected at a radius of 500 meters from the point (x,y). The sensor feature may include (but are not limited to): vehicle speed (average values calculated for either direction and are recorded separately), heading degree (average values are calculated for either direction and are recorded separately), friction (braking pattern)—slippery road condition, windshield wiper (for example, to detect if the road is wet due to rain) weather condition, fog lights—ON/OFF (for example, to detect if there is low visibility due to fog or other weather condition, anti-brake system data (ABS), for example that related to a slippery road condition, an average number of lane changes in a time period, congestion (number of vehicle on a segment of, for example, 100 meters), among others. The model is trained on thousands of such locations with the sensor data and traffic violation data. In application, when a vehicle approaches a junction or a traffic light (a potential traffic violation hot-spot), the model collects the vehicle sensor data of all the vehicles and feed the input data to the trained model. The output of the model is a probability (in the range of 0 to 1) of the location being a traffic violation hotspot. If the probability is greater than, for example, 75% then the location is flagged as a traffic violation hotspot.

(31) In an embodiment, a combination that leverages map data (e.g., static features) with more dynamic and contextual data (captured by vehicles on the road near the areas of interest), is used. The mapping system **121** combines the probability provided by a map feature model and a sensor data model with a simple weighted average. In an example both models may carry an equal weight of 0.5. For example, if a map feature model gives the output as 0.8 and a sensor data model gives the output as 0.75 then the combination will calculate the combined probability as  $0.5*0.8+0.5*0.75=0.775$ .

(32) Different probabilities and thresholds may be used for determining when an area or location is a hotspot. Probabilities and thresholds may, for example, be relative for an area compared to other nearby areas. For example, in a region with a low level of traffic violations, a certain location may still have a low probability of traffic violations but still be very high compared to other locations. Conversely, in a region with a high level of traffic violations, every area or location may pass a threshold for the low-level area and as such a higher threshold may be used to differentiate locations from one another and where they sit on a ranking of traffic violation hotspots.

(33) In an embodiment, the models are trained using machine learning techniques. The models may be, for example, classifiers that are trained using supervised learning. The models may classify, based on the input sensor or map feature data, whether or not an area is a hotspot for traffic violations (or individually for each different violation). The model(s) may include a neural network that is defined as a plurality of sequential feature units or layers. Sequential is used to indicate the general flow of output feature values from one layer to input to a next layer. Sequential is used to indicate the general flow of output feature values from one layer to input to a next layer. The information from the next layer is fed to a next layer, and so on until the final output. The layers may only feed forward or may be bi-directional, including some feedback to a previous layer. The nodes of each layer or unit may connect with all or only a sub-set of nodes of a previous and/or subsequent layer or unit. Skip connections may be used, such as a layer outputting to the sequentially next layer as well as other layers. Rather than pre-programming the features and trying to relate the features to attributes, the deep architecture is defined to learn the features at different levels of abstraction based on the input data. The features are learned to reconstruct lower-level features (i.e., features at a more abstract or compressed level). Each node of the unit represents a feature. Different units are provided for learning different features. Various units or layers may be used, such as convolutional, pooling (e.g., max pooling), deconvolutional, fully connected, or other types of layers. Within a unit or layer, any number of nodes is provided. For example, 100 nodes are provided. Later or subsequent units may have more, fewer, or the same number of nodes.

(34) Unsupervised learning may be used to compute classification, based on the distribution of the

samples, using methods such as k-nearest neighbor. In supervised learning, the classification step may happen in the last layer, and takes the key features of the sample as input from the previous layers. There are different classification functions, depending on the use case. An embodiment uses a Softmax function—where for each sample, the result is the probability distribution over the classes.

(35) Different neural network configurations and workflows may be used for the network such as a convolution neural network (CNN), deep belief nets (DBN), or other deep networks. CNN learns feed-forward mapping functions while DBN learns a generative model of data. In addition, CNN uses shared weights for all local regions while DBN is a fully connected network (e.g., including different weights for all regions of a feature map. The training of CNN is entirely discriminative through backpropagation. DBN, on the other hand, employs the layer-wise unsupervised training (e.g., pre-training) followed by the discriminative refinement with backpropagation if necessary. In an embodiment, the arrangement of the trained network is a fully convolutional network (FCN). Alternative network arrangements may be used, for example, a 3D Very Deep Convolutional Networks (3D-VGGNet). VGGNet stacks many layer blocks containing narrow convolutional layers followed by max pooling layers. A 3D Deep Residual Networks (3D-ResNet) architecture may be used. A Resnet uses residual blocks and skip connections to learn residual mapping.

(36) The training data for the model/network (and other networks) includes ground truth data or gold standard data, for example actual detected or identified traffic violation data. Ground truth data and gold standard data is data that includes correct or reasonably accurate labels that are verified manually or by some other accurate method. The training data may be acquired at any point prior to inputting the training data into the network. In an example operation, one or more of the models are configured as classifiers. The network inputs the training data (e.g., sensor data, features data, or sensor and feature data) and outputs a prediction. The prediction is compared to the annotations from the training data. A loss function may be used to identify the errors from the comparison. The loss function serves as a measurement of how far the current set of predictions are from the corresponding true values. Some examples of loss functions that may be used include Mean-Squared-Error, Root-Mean-Squared-Error, and Cross-entropy loss. Mean Squared Error loss, or MSE for short, is calculated as the average of the squared differences between the predicted and actual values. Root-Mean Squared Error is similarly calculated as the average of the root squared differences between the predicted and actual values. For cross-entropy loss each predicted probability is compared to the actual class output value (0 or 1) and a score is calculated that penalizes the probability based on the distance from the expected value. The penalty may be logarithmic, offering a small score for small differences (0.1 or 0.2) and enormous score for a large difference (0.9 or 1.0). During training and over repeated iterations, the network attempts to minimize the loss function as the result of a lower error between the actual and the predicted values means the network has done a good job in learning. Different optimization algorithms may be used to minimize the loss function, such as, for example, gradient descent, Stochastic gradient descent, Batch gradient descent, Mini-Batch gradient descent, among others. The process of inputting, outputting, comparing, and adjusting is repeated for a predetermined number of iterations with the goal of minimizing the loss function.

(37) One adjusted and trained, the model(s) are configured to calculate traffic violation hotspot predictions. In an embodiment, the traffic violation probabilities may be published in a map update. For example, if the mean of the traffic violation for a given area and for given hour is same (or less) as that of the nation (or county or state) mean then the location is not flagged. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 15% then it is flagged as mild traffic violation zone. This region may be shown as amber color on the map. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 25% then it is flagged as moderate traffic violation zone. This region may be

shown as light red color on a map. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 50%+ then it is flagged as high traffic violation zone. This region may be shown as dark red color on the map.

(38) Table 1 below shows some examples of how a location is depicted based on the hour of the day.

(39) TABLE-US-00001 % OF RISK AS MEAN # OF NATIONAL NATIONAL SHOWN ON HOUR EVENTS AVERAGE AVERAGE MAP 6 am to 7 am 15 16 -6% No flag 7 am to 8 am 21 22 -4% No flag 8 am to 9 am 30 28 7% Amber flag 9 am to 10 am 50 43 16% Light red flag 10 am to 11 am 45 43 5% Amber flag 11 am to 12 am 25 28 -10% No flag

(40) FIG. 2 depicts a map of an intersection that can be displayed with different colors or annotations depending on the risk for a respective time. As depicted, the intersections are highlighted/shaded differently based on a probability of a traffic violation. Intersections with higher than mean values may be indicated on a map using different colors/annotations/shading etc. As depicted the map includes several different risk levels for different intersections ranging from a 2% decrease to a 35% increase in a risk of a traffic violation. The map display also depicts a suggested route (in the map display as an arrow) that takes into account the different probabilities of a traffic violation. As shown, the route avoids the more at-risk intersections. The traffic violation probabilities may not be definitive but, for example, when a choice between a first route and a second route is identified that otherwise is equal time and distance wise, the lower risk route may be recommended. Different types of displays or interfaces may be used, for example, each road segment may be colored or shaded as opposed to intersections or nodes. Only the highest risk areas may be displayed or recent increases or spikes in the probability.

(41) In an embodiment, during application, the model(s) are used to determine the probability of traffic violation happening at a given location. If the probability is less than a threshold, then the device 122/vehicle may ignore the warning and the vehicle may continue on. If the probability is greater than a threshold, the vehicle determines that the threat from an accident, due to traffic violation hotspot, is valid. When the traffic violation probability is determined to be greater than the threshold then based on the location and time of the vehicle, feature data for the location/region is acquired and input into a trained model that is configured to predict the probability of a traffic accident based on the acquired feature data and the traffic violation data. The feature data may include, for example, data related to an urban/rural flag (from the geographic database 123), population density (online data), presence of a logistic company within 1 km radius (pizza shop, warehouse, courier company, (the existence of these companies nearby may for example, increase the threat of an accident due to delivery vehicle), the season/time quartile, and historic accident event that has taken place for a given set of features. Using historical data, a machine learning model is trained and configured based on this data set that is configured to output the probability of an accident happening at a particular location and at a specific level of traffic violation spot.

(42) Similar to the models/networks for determining traffic violation hotspots, the model for determining the probability of an accident may be configured as a classifier. Classification is a form of data analysis which takes each instance of a data set and assigns it to particular class. A classification-based network attempts to classify locations as either normal or hotspots. The challenge of classification is to reduce the number of false positives (detection of a normal location as abnormal) and false negatives (detection of a hotspot as normal). Different techniques may be used to speed up the process or provide more accurate results. Feature selection, for example, may be used to improve performance through the removal of redundant or irrelevant attributes. Feature selection methods generate a new set of attributes by selecting only a subset of the original attributes.

(43) Supervised or unsupervised learning may be used. Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs. It

infers a function from labeled training data consisting of a set of training examples. Unsupervised learning algorithms lack individual target variables and instead have the goal of characterizing a data set in general. Unsupervised machine learning algorithms are often used to group (cluster) data sets, e.g., to identify relationships between individual data points (that may include of any number of attributes) and group them into clusters. Different algorithms or techniques may be used such as clustering. Clustering is the process of partitioning data into groups according to certain characteristics of data. Clustering splits data into groups of similar objects. Every group, called cluster, includes of members that are quite similar and members from the various clusters are different from each other.

(44) In an embodiment there are two layers of model running in sequence. The first model (determine the traffic violation hotspot) is running continuously, for example, every 0.5 Km. If the probability of traffic violation hotspot is greater than a threshold only then is the second layer of the accident probability model is activated. In doing so the vehicle system is not burdened with continuous data processing. The proposed solution is a system that allows users to benefit from increased safety by dynamically predicting the areas where traffic violations are most likely to occur. The user may be a vehicle, vehicle driver or Traffic management department.

(45) The predictions may be used for several applications, for example, where authorities want to understand where such traffic violations are likely to happen so that they can act proactively by deploying police forces in the area or preparing ambulances or other emergency responses team in case of accidents. In addition, being able to predict that some intersections are likely to see traffic violations before they even happen can help city planners do a better job in designing new intersections or planning road works and updates. Such learnings made by the proposed models may then be embedded in urban mobility modelling software so that everyone can benefit from it. One key advantage of such a prediction function is that a city does not need to gather lots of historical information in that place before being able to act. The city or organization may already act based on the learnings that have been done in areas which have similar characteristics as described above. This reduces the time to apply measures and helps save lives. Further benefits include increased safety, cheaper simulations and deployment, better predictability, traffic management, better city planning, and better data for businesses such as insurance companies.

(46) The proposed systems and methods develop a predictor based on real-time information related to traffic and traffic incidents. In an embodiment, a warning system provides details of traffic violations to a device **122** as the device **122**/vehicle traverses the roadway. For example, a warning may be displayed using a navigation application such as “INCREASED RISK” for a location indicating that there is a greater than the national average risk of wrong way driving taking place on this segment of the roadway.

(47) FIG. **3** illustrates an example mobile device **122** for the system of FIG. **1** that calculates a probability of a traffic violation and/or traffic accident based on sensor and/or feature data about the roadway. The mobile device **122** may include a bus **910** that facilitates communication between a controller **900** that may be implemented by a processor **901** and/or an application specific controller **902**, which may be referred to individually or collectively as controller **900**, and one or more other components including a database **903**, a memory **904**, a computer readable medium **905**, a communication interface **918**, a radio **909**, a display **914**, a camera **915**, a user input device **916**, position circuitry **922**, ranging circuitry **923**, and vehicle circuitry **924**. The contents of the database **903** are described with respect to the geographic database **123**. The device-side database **903** may be a user database that receives data in portions from the database **903** of the mobile device **122**. The communication interface **918** connected to the internet and/or other networks (e.g., network **127** shown in FIG. **1**). The vehicle circuitry **924** may include any of the circuitry and/or devices described with respect to FIG. **8**. Additional, different, or fewer components may be included.

(48) FIG. **4** depicts an example workflow for calculating a probability of a traffic violation and/or

traffic accident based on sensor and/or feature data about the roadway using the device **122** of FIG. **3**. As presented in the following sections, the acts may also in part be performed using any combination of the components indicated in FIG. **1**, FIG. **3**, or FIG. **8**. For example, certain acts may be performed by the server **125**, the device **122**, the mapping system **121**, or a combination thereof. Additional, different, or fewer acts may be provided. The acts are performed in the order shown or other orders. The acts may also be repeated. Certain acts may be skipped.

(49) The mobile device **122** may be a personal navigation device (“PND”), a portable navigation device, a mobile phone, a personal digital assistant (“PDA”), a watch, a tablet computer, a notebook computer, and/or any other known or later developed mobile device or personal computer. The mobile device **122** may also be an automobile head unit, infotainment system, and/or any other known or later developed automotive navigation system. Non-limiting embodiments of navigation devices may also include relational database service devices, mobile phone devices, car navigation devices, and navigation devices used for air or water travel.

(50) At act **A110**, the device **122** acquires data related to a location of a roadway. The data may be map-matched to a road segment or node. The device **122** is configured to determine its location and, for example, an upcoming location using the position circuitry **922**, ranging circuitry **923**, vehicle circuitry **924**, and the geographic database **123**. The positioning circuitry **922** may include suitable sensing devices that measure the traveling distance, speed, direction, and so on, of the mobile device **122**. The positioning system may also include a receiver and correlation chip to obtain a GPS signal. Alternatively, or additionally, the one or more detectors or sensors may include an accelerometer and/or a magnetic sensor built or embedded into or within the interior of the mobile device **122**. The accelerometer is operable to detect, recognize, or measure the rate of change of translational and/or rotational movement of the mobile device **122**. The magnetic sensor, or a compass, is configured to generate data indicative of a heading of the mobile device **122**. Data from the accelerometer and the magnetic sensor may indicate orientation of the mobile device **122**. The mobile device **122** receives location data from the positioning system. The location data indicates the location of the mobile device **122**.

(51) The positioning circuitry **922** may include a Global Positioning System (GPS), Global Navigation Satellite System (GLONASS), or a cellular or similar position sensor for providing location data. The positioning system may utilize GPS-type technology, a dead reckoning-type system, cellular location, or combinations of these or other systems. The positioning circuitry **922** may include suitable sensing devices that measure the traveling distance, speed, direction, and so on, of the mobile device **122**. The positioning system may also include a receiver and correlation chip to obtain a GPS signal. The mobile device **122** receives location data from the positioning system. The location data indicates the location of the mobile device **122**.

(52) The position circuitry **922** may also include gyroscopes, accelerometers, magnetometers, or any other device for tracking or determining movement of a mobile device **122**. The gyroscope is operable to detect, recognize, or measure the current orientation, or changes in orientation, of a mobile device **122**. Gyroscope orientation change detection may operate as a measure of yaw, pitch, or roll of the mobile device **122**.

(53) The device **122** is configured to acquire the data for the upcoming location using one or more sensors and/or the geographic database **123**. The one or more sensors may include ranging circuitry **923**, image/video cameras, weather sensors, occupant sensors, and any other vehicle sensor that collects data about the vehicle or the environment around the vehicle. For example, the ranging circuitry **923** may include a LIDAR system, a RADAR system, a structured light camera system, SONAR, or any device configured to detect the range or distance to objects from the mobile device **122**. The controller **900** of the device **122** may also communicate with a vehicle engine control unit (ECU) that operates one or more driving mechanisms (e.g., accelerator, brakes, steering device). Alternatively, the mobile device **122** may be the vehicle ECU that operates the one or more driving mechanisms directly.

(54) At act A120, the device 122 generates, using at least one model trained using machine learning, a probability score on how probable is the location is likely to be a traffic violation hotspot. The at least one model may be stored in the memory 904. The memory 904 may be a volatile memory or a non-volatile memory. The memory 904 may include one or more of a read only memory (ROM), random access memory (RAM), a flash memory, an electronic erasable program read only memory (EEPROM), or other type of memory. The memory 904 may be removable from the mobile device 122, such as a secure digital (SD) memory card. The at least one model may be trained, configured, and updated at the mapping system 121. The mapping system 121 may iteratively train or configure the model using a set of historical training data that includes annotated (known or identified) traffic violation events. The training data is input into the model which outputs a prediction (probability). The output is compared the annotation. The comparison is used to adjust the model/network until the model is optimized. For the machine learning task described above and herein, the model (also referred to as machine learning model, neural network, or network) may be trained using one or more optimization algorithms such as gradient decent. Training using an optimization method such as gradient descent includes determining how close the model estimates the target function. The determination may be calculated a number of different ways that may be specific to the particular model being trained. The cost function involves evaluating the parameters in the model by calculating a prediction for the model for each training instance in the dataset and comparing the predictions to the actual output values and calculating an average error value (such as a value of squared residuals or SSR in the case of linear regression). In a simple example of linear regression, a line is fit to a set of points. An error function (also called a cost function) is defined that measures how good (accurate) a given line is. In an example, the function inputs the points and return an error value based on how well the line fits the data. To compute the error for a given line, in this example, each point (x, y) is iterated in the data set and the sum the square distances between each point's y value and the candidate line's y value is calculated as the error function.

(55) Gradient descent is used to minimize the error functions. Given a function defined by a set of parameters, gradient descent starts with an initial set of parameter values and iteratively moves toward a set of parameter values that minimize the function. The iterative minimization is based on a function that takes steps in the negative direction of the function gradient. A search for minimizing parameters starts at any point and allows the gradient descent algorithm to proceed downhill on the error function towards a best outcome. Each iteration updates the parameters that yield a slightly different error than the previous iteration. A learning rate variable is defined that controls how large of a step that is taken downhill during each iteration.

(56) Alternative optimization algorithms may be used. For example, stochastic gradient decent is a variation of gradient decent that may be used. Nesterov accelerated gradient (NAG) is another algorithm that solves a problem of momentum when an algorithm reaches the minima i.e., the lowest point on the curve. Adaptive Moment Estimation (Adam) is another method that computes adaptive learning rates for each parameter. In addition to storing an exponentially decaying average of past squared gradients like AdaDelta, Adam also keeps an exponentially decaying average of past gradients  $M(t)$ , similar to momentum. Depending on the model, different types of optimization algorithms, e.g., first order or second order (hessian) may be used. Any algorithm that executes iteratively by comparing various solutions until an optimum or a satisfactory solution is found may be used to train the model.

(57) The device 122 may communicate with the mapping system 121 to receive updates for the at least one model. The communication interface 918 and/or communication interface 918 may include any operable connection. An operable connection may be one in which signals, physical communications, and/or logical communications may be sent and/or received. An operable connection may include a physical interface, an electrical interface, and/or a data interface. The communication interface 918 provides for wireless and/or wired communications in any now

known or later developed format. The radio **909** may be configured to radio frequency communication (e.g., generate, transit, and receive radio signals) for any of the wireless networks described herein including cellular networks, the family of protocols known as WIFI or IEEE 802.11, the family of protocols known as Bluetooth, or another protocol.

(58) In an embodiment, the data related to the upcoming location comprises map feature data, wherein the at least one model is trained to learn which map features are the most relevant based on the correlations between those features and the traffic violations, wherein once those map features are identified and weighted in the at least one model, the at least one model is configured to determine the generate the probability score based on the acquired map feature data. The map feature data may be acquired in act **A110** from the geographic database **123**. The geographic database **123** includes information about one or more geographic regions. FIG. 5 illustrates a map of a geographic region **202**. The geographic region **202** may correspond to a metropolitan or rural area, a state, a country, or combinations thereof, or any other area. Located in the geographic region **202** are physical geographic features, such as roads, points of interest (including businesses, municipal facilities, etc.), lakes, rivers, railroads, municipalities, etc.

(59) FIG. 5 further depicts an enlarged map **204** of a portion **206** of the geographic region **202**. The enlarged map **204** illustrates part of a road network **208** in the geographic region **202**. The road network **208** includes, among other things, roads and intersections located in the geographic region **202**. As shown in the portion **206**, each road in the geographic region **202** is composed of one or more road segments **210**. A road segment **210** represents a portion of the road. Road segments **210** may also be referred to as links. Each road segment **210** is shown to have associated with it one or more nodes **212**; one node represents the point at one end of the road segment and the other node represents the point at the other end of the road segment. The node **212** at either end of a road segment **210** may correspond to a location at which the road meets another road, i.e., an intersection, or where the road dead ends.

(60) As depicted in FIG. 6, in one embodiment, the geographic database **123** contains geographic data **302** that represents some of the geographic features in the geographic region **202** depicted in FIG. 5. The data **302** contained in the geographic database **123** may include data that represent the road network **208**. In FIG. 4, the geographic database **123** that represents the geographic region **202** may contain at least one road segment database record **304** (also referred to as “entity” or “entry”) for each road segment **210** in the geographic region **202**. The geographic database **123** that represents the geographic region **202** may also include a node database record **306** (or “entity” or “entry”) for each node **212** in the geographic region **202**. The terms “nodes” and “segments” represent only one terminology for describing these physical geographic features, and other terminology for describing these features is intended to be encompassed within the scope of these concepts.

(61) The geographic database **123** may include feature data **308-312**. The feature data **312** may represent types of geographic features. For example, the feature data may include roadway data **308** including signage data, lane data, traffic signal data, physical and painted features like dividers, lane divider markings, road edges, center of intersection, stop bars, overpasses, overhead bridges, etc. The roadway data **308** may be further stored in sub-indices that account for different types of roads or features. The point of interest data **310** may include data or sub-indices or layers for different types of points of interest. The point of interest data may include point of interest records comprising a type (e.g., the type of point of interest, such as restaurant, fuel station, hotel, city hall, police station, historical marker, ATM, golf course, truck stop, vehicle chain-up stations, etc.), location of the point of interest, a phone number, hours of operation, etc. The feature data **312** may include other roadway features.

(62) The geographic database **123** also includes indexes **314**. The indexes **314** may include various types of indexes that relate the different types of data to each other or that relate to other aspects of the data contained in the geographic database **123**. For example, the indexes **314** may relate the

nodes in the node data records **306** with the end points of a road segment in the road segment data records **304**.

(63) FIG. **6** shows some of the components of a road segment data record **304** contained in the geographic database **123** according to one embodiment. The road segment data record **304** may include a segment ID **304(1)** by which the data record can be identified in the geographic database **123**. Each road segment data record **304** may have associated information such as “attributes”, “fields”, etc. that describes features of the represented road segment. The road segment data record **304** may include data **304(2)** that indicate the restrictions, if any, on the direction of vehicular travel permitted on the represented road segment. The road segment data record **304** may include data **304(3)** that indicate a speed limit or speed category (i.e., the maximum permitted vehicular speed of travel) on the represented road segment. The road segment data record **304** may also include classification data **304(4)** indicating whether the represented road segment is part of a controlled access road (such as an expressway), a ramp to a controlled access road, a bridge, a tunnel, a toll road, a ferry, and so on. The road segment data record **304** may include data **304(5)** related to points of interest. The road segment data record **304** may include data **304(6)** that describes lane configurations. The road segment data record **304** also includes data **304(7)** providing the geographic coordinates (e.g., the latitude and longitude) of the end points of the represented road segment. In one embodiment, the data **304(7)** are references to the node data records **306** that represent the nodes corresponding to the end points of the represented road segment. The road segment data record **304** may also include or be associated with other data **304(7)** that refer to various other attributes of the represented road segment such as coordinate data for shape points, POIs, signage, other parts of the road segment, etc. The various attributes associated with a road segment may be included in a single road segment record, or may be included in more than one type of record which cross-references each other. For example, the road segment data record **304** may include data identifying what turn restrictions exist at each of the nodes which correspond to intersections at the ends of the road portion represented by the road segment, the name or names by which the represented road segment is known, the street address ranges along the represented road segment, and so on.

(64) FIG. **7** also shows some of the components of a node data record **306** which may be contained in the geographic database **123**. Each of the node data records **306** may have associated information (such as “attributes”, “fields”, etc.) that allows identification of the road segment(s) that connect to it and/or a geographic position (e.g., latitude and longitude coordinates). For the embodiment shown in FIG. **7**, the node data records **306(1)** and **306(2)** include the latitude and longitude coordinates **306(1)(1)** and **306(2)(1)** for their node. The node data records **306(1)** and **306(2)** may also include other data **306(1)(3)** and **306(2)(3)** that refer to various other attributes of the nodes.

(65) The data in the geographic database **123** may be organized using a graph that specifies relationships between entities. A location graph is a graph that includes relationships between location objects in a variety of ways. Objects and their relationships may be described using a set of labels. Objects may be referred to as “nodes” of the location graph, where the nodes and relationships among nodes may have data attributes. The organization of the location graph may be defined by a data scheme that defines the structure of the data. The organization of the nodes and relationships may be stored in an ontology which defines a set of concepts where the focus is on the meaning and shared understanding. These descriptions permit mapping of concepts from one domain to another. The ontology is modeled in a formal knowledge representation language which supports inferencing and is readily available from both open-source and proprietary tools.

(66) In an embodiment, the data related to the upcoming location comprises sensor data acquired by a vehicle, wherein the at least one model is trained to input sensor data and output a probability that a location is traffic violation hotspot.

(67) At act **A130**, the device **122** determines, that the probability score exceeds a threshold score. The threshold score may be set by the device **122**, an operator, the mapping system **121** or by other



means. The threshold may be region specific or may be based on the type of vehicle that is being driven. Different fleets or organizations may have different standards for safety. Different autonomous vehicles may also be better equipped to handle certain circumstances.

(68) In an embodiment, the device **122** may publish or display the probability score for traffic violations. For example, if the mean of the traffic violation for a given area and for given hour is same (or less) as that of the nation (or county or state) mean then the location is not flagged. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 15% then it is flagged as mild traffic violation zone. This region is shown as amber color on the map. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 25% then it is flagged as moderate traffic violation zone. This region is shown as light red color on map. If the mean of the traffic violation for a given area and for given hour is greater than that of the nation (or county or state) mean then the location (or area) by 50%+ then it is flagged as high traffic violation zone. This region is shown as dark red color on the map.

(69) At act **A140**, the device **122** acquires feature data about the location and a region encompassing the upcoming location. The feature data may include, for example, data related to an urban/rural flag (from the geographic database **123**), population density (online data), presence of a logistic company within 1 km radius (pizza shop, warehouse, courier company, (the existence of these companies nearby may, for example, increase the threat of an accident due to delivery vehicle), the season/time quartile, and historic accident event that has taken place for a given set of features. The feature data may be stored in the geographic database **123**, for example locally, or may be acquired from the mapping system **121**.

(70) At act **A150**, the device **122** generates using a machine trained model, a probability of an accident happening at the location. Using historical data, a machine learning model is trained and configured by the mapping system **121** based on this data set. The machine learning model is configured to output the probability of an accident happening at a particular location and at a specific level of traffic violation spot.

(71) At act **A160**, the device **122** generates and transmits an alert for the location based on the probability calculated by the machine learning model. The alert may be an update to a displayed map. The alert may be, for example a routing instruction to take a different route. The routing instructions may be provided by display **914**. The mobile device **122** may be configured to execute routing algorithms to determine an optimum route to travel along a road network from an origin location to a destination location in a geographic region. Using input(s) including map matching values from the server **125**, a mobile device **122** examines potential routes between the origin location and the destination location to determine the optimum route. The mobile device **122**, which may be referred to as a navigation device, may then provide the end user with information about the optimum route in the form of guidance that identifies the maneuvers required to be taken by the end user to travel from the origin to the destination location. Some mobile devices **122** show detailed maps on displays outlining the route, the types of maneuvers to be taken at various locations along the route, locations of certain types of features, and so on. Possible routes may be calculated based on a Dijkstra method, an A-star algorithm or search, and/or other route exploration or calculation algorithms that may be modified to take into consideration assigned cost values of the underlying road segments.

(72) A user may interact with the map/navigation system/alert using an input device **916**. The input device **916** may be one or more buttons, keypad, keyboard, mouse, stylus pen, trackball, rocker switch, touch pad, voice recognition circuit, or other device or component for inputting data to the mobile device **122**. The input device **916** and display **914** may be combined as a touch screen, which may be capacitive or resistive. The display **914** may be a liquid crystal display (LCD) panel, light emitting diode (LED) screen, thin film transistor screen, or another type of display. The output interface of the display **914** may also include audio capabilities, or speakers. In an embodiment, the

input device **916** may involve a device having velocity detecting abilities.

(73) The alert and probability information may be used to be aware of such risks or avoid those hotspots. In an example, when the vehicle is approaching the traffic violation hotspot, the device **122** might prompt the user to take over the control of the vehicle. The controller **900** may reduce speed or behaviors in such areas. In a practical example, vehicles may drive in the opposite lane (overtaking) to go over accidents which happened on their lane. If an accident is detected in a given area on a given lane, it increases the risk of people using the opposite lane to go over it (depending on the area, validated by probe data in that area). Based on this information, the device **122** might predict that when an accident happens in the future on that same lane and conditions are similar (traffic, weather, time of day, etc), then there will likely be people attempting to overtake on the opposite lane, leading to increased safety risks. If there is long congestion on one side of the road and the other side of the road has a sparse vehicle then the motorist may have the tendency to drive from the wrong side of the road and beat the congestion. Similarly, in a commercial area (where there are offices) there may be higher cases of traffic light jumping during peak hours (probable reason—getting late for work). Such patterns can be identified from this traffic violation hotspot. Identification of predictors can help in reducing traffic related risks.

(74) In response to the alert, a vehicle or driver may decide to take a different route if the dynamically computed risk is over a given threshold. Vehicles in both directions may be informed of the increased risk for a specific time period. Pedestrians may be informed that dangerous driving is more likely to occur when a vehicle is stopped on a street with given characteristics (e.g., one driving lane in each direction). Police/assistance may be notified to come and support faster the incident that occurred in such areas with higher associated risk. Police or emergency services may also preemptively come and control to prevent such dangerous behaviors in a proactive way thanks to the prediction capability.

(75) In an embodiment, the device **122** may alert or otherwise provide instructions for an autonomous vehicle to perform a maneuver. FIG. 8 illustrates an exemplary vehicle **124** for providing location-based services, navigation services, or applications using the systems and methods described herein as well as collecting data for such services or applications described herein. The vehicles **124** may include a variety of devices that collect position data as well as other related sensor data for the surroundings of the vehicle **124**. The position data may be generated by a global positioning system, a dead reckoning-type system, cellular location system, or combinations of these or other systems, which may be referred to as position circuitry or a position detector. The positioning circuitry may include suitable sensing devices that measure the traveling distance, speed, direction, and so on, of the vehicle **124**. The positioning system may also include a receiver and correlation chip to obtain a GPS or GNSS signal. Alternatively, or additionally, the one or more detectors or sensors may include an accelerometer built or embedded into or within the interior of the vehicle **124**. The vehicle **124** may include one or more distance data detection device or sensor, such as a LIDAR device. The distance data detection sensor may include a laser range finder that rotates a mirror directing a laser to the surroundings or vicinity of the collection vehicle on a roadway or another collection device on any type of pathway.

(76) A connected vehicle includes a communication device and an environment sensor array for reporting the surroundings of the vehicle **124** to the server **125**. The connected vehicle may include an integrated communication device coupled with an in-dash navigation system. The connected vehicle may include an ad-hoc communication device such as a mobile device **122** or smartphone in communication with a vehicle system. The communication device connects the vehicle to a network including at least one other vehicle and at least one server **125**. The network may be the Internet or connected to the internet.

(77) The sensor array may include one or more sensors configured to detect surroundings of the vehicle **124**. The sensor array may include multiple sensors. Example sensors include an optical distance system such as LiDAR **956**, an image capture system **955** such as a camera, a sound

distance system such as sound navigation and ranging (SONAR), a radio distancing system such as radio detection and ranging (RADAR) or another sensor. The camera may be a visible spectrum camera, an infrared camera, an ultraviolet camera, or another camera.

(78) In some alternatives, additional sensors may be included in the vehicle **124**. An engine sensor **951** may include a throttle sensor that measures a position of a throttle of the engine or a position of an accelerator pedal, a brake sensor that measures a position of a braking mechanism or a brake pedal, or a speed sensor that measures a speed of the engine or a speed of the vehicle wheels. Another additional example, vehicle sensor **953**, may include a steering wheel angle sensor, a speedometer sensor, or a tachometer sensor.

(79) A mobile device **122** may be integrated in the vehicle **124**, which may include assisted driving vehicles such as autonomous vehicles, highly assisted driving (HAD), and advanced driving assistance systems (ADAS). Any of these assisted driving systems may be incorporated into mobile device **122**. Alternatively, an assisted driving device may be included in the vehicle **124**. The assisted driving device may include memory, a processor, and systems to communicate with the mobile device **122**. The assisted driving vehicles may respond to the lane marking indicators (lane marking type, lane marking intensity, lane marking color, lane marking offset, lane marking width, or other characteristics) received from geographic database **123** and the server **125** and driving commands or navigation commands.

(80) The term autonomous vehicle may refer to a self-driving or driverless mode in which no passengers are required to be on board to operate the vehicle. An autonomous vehicle may be referred to as a robot vehicle or an automated vehicle. The autonomous vehicle may include passengers, but no driver is necessary. These autonomous vehicles may park themselves or move cargo between locations without a human operator. Autonomous vehicles may include multiple modes and transition between the modes. The autonomous vehicle may steer, brake, or accelerate the vehicle based on the position of the vehicle in order, and may respond to lane marking indicators (lane marking type, lane marking intensity, lane marking color, lane marking offset, lane marking width, or other characteristics) received from geographic database **123** and the server **125** and driving commands or navigation commands.

(81) A highly assisted driving (HAD) vehicle may refer to a vehicle that does not completely replace the human operator. Instead, in a highly assisted driving mode, the vehicle may perform some driving functions and the human operator may perform some driving functions. Vehicles may also be driven in a manual mode in which the human operator exercises a degree of control over the movement of the vehicle. The vehicles may also include a completely driverless mode. Other levels of automation are possible. The HAD vehicle may control the vehicle through steering or braking in response to the on the position of the vehicle and may respond to lane marking indicators (lane marking type, lane marking intensity, lane marking color, lane marking offset, lane marking width, or other characteristics) received from geographic database **123** and the server **125** and driving commands or navigation commands.

(82) Similarly, ADAS vehicles include one or more partially automated systems in which the vehicle alerts the driver. The features are designed to avoid collisions automatically. Features may include adaptive cruise control, automate braking, or steering adjustments to keep the driver in the correct lane. ADAS vehicles may issue warnings for the driver based on the position of the vehicle or based on the lane marking indicators (lane marking type, lane marking intensity, lane marking color, lane marking offset, lane marking width, or other characteristics) received from geographic database **123** and the server **125** and driving commands or navigation commands.

(83) The term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(84) In a particular non-limiting, embodiment, the computer-readable medium may include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium may be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium may include a magneto-optical or optical medium, such as a disk or tapes or other storage device to capture carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

(85) In an alternative embodiment, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, may be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments may broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that may be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

(86) In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited embodiment, implementations may include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing may be constructed to implement one or more of the methods or functionalities as described herein.

(87) Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the invention is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP, HTTPS) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

(88) A computer program (also known as a program, software, software application, script, or code) may be written in any form of programming language, including compiled or interpreted languages, and it may be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program may be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program may be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(89) The processes and logic flows described in the specification may be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows may also be performed by, and apparatus may also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

(90) As used in the application, the term 'circuitry' or 'circuit' refers to all of the following: (a)

hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry) and (b) to combinations of circuits and software (and/or firmware), such as (as applicable): (i) to a combination of processor(s) or (ii) to portions of processor(s)/software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions) and (c) to circuits, such as a microprocessor(s) or a portion of a microprocessor(s), that require software or firmware for operation, even if the software or firmware is not physically present.

(91) This definition of ‘circuitry’ applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term “circuitry” would also cover an implementation of merely a processor (or multiple processors) or portion of a processor and its (or their) accompanying software and/or firmware. The term “circuitry” would also cover, for example and if applicable to the particular claim element, a baseband integrated circuit or applications processor integrated circuit for a mobile phone or a similar integrated circuit in server, a cellular network device, or other network device.

(92) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and anyone or more processors of any kind of digital computer. Generally, a processor receives instructions and data from a read only memory or a random-access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data.

Generally, a computer also includes, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer may be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a GPS receiver, to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media, and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The memory may be a non-transitory medium such as a ROM, RAM, flash memory, etc. The processor and the memory may be supplemented by, or incorporated in, special purpose logic circuitry.

(93) To provide for interaction with a user, embodiments of the subject matter described in this specification may be implemented on a device having a display, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices may be used to provide for interaction with a user as well; for example, feedback provided to the user may be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including acoustic, speech, or tactile input.

(94) Embodiments of the subject matter described in this specification may be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user may interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system may be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), e.g., the Internet.

(95) The computing system may include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and

having a client-server relationship to each other.

(96) The illustrations of the embodiments described herein are intended to provide a general understanding of the structure of the various embodiments. The illustrations are not intended to serve as a complete description of all of the elements and features of apparatus and systems that utilize the structures or methods described herein. Many other embodiments may be apparent to those of skill in the art upon reviewing the disclosure. Other embodiments may be utilized and derived from the disclosure, such that structural and logical substitutions and changes may be made without departing from the scope of the disclosure. Additionally, the illustrations are merely representational and may not be drawn to scale. Certain proportions within the illustrations may be exaggerated, while other proportions may be minimized. Accordingly, the disclosure and the figures are to be regarded as illustrative rather than restrictive.

(97) While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments may also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment may also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(98) Similarly, while operations are depicted in the drawings and described herein in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a single software product or packaged into multiple software products.

(99) One or more embodiments of the disclosure may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any particular invention or inventive concept. Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are apparent to those of skill in the art upon reviewing the description.

(100) The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72(b) and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

(101) It is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is understood that the following claims including all equivalents are intended to define the scope of the invention. The claims should not be read as limited to the described order or

elements unless stated to that effect. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the invention.

## Claims

1. A method comprising: acquiring data related to a location of a roadway, wherein the data related to the location comprises map feature data and sensor data acquired by a vehicle; generating, using at least one model trained using machine learning, a probability score on how probable the location is to be a traffic violation hotspot based on the acquired data, wherein the at least one model comprises a first model and a second model, wherein the first model is trained to learn which map features are the most relevant based on correlations between those map features and traffic violations, wherein once those map features are identified and weighted in the at least one model, the at least one model is configured to generate a first probability score based on the acquired map feature data, wherein the second model is trained to input sensor data and output a second probability score that the location is traffic violation hotspot, wherein the probability score is calculated as a weighted average of the first probability score and the second probability score; determining, that the probability score exceeds a threshold score; acquiring feature data about the location and a region encompassing the location; generating, by a machine learning model, a probability of an accident occurring at the location; and providing one or more instructions for a vehicle to perform a maneuver for the location based on the probability.
2. The method of claim 1, wherein the data related to the location comprises map feature data, wherein the at least one model is trained to learn which map features are the most relevant based on the correlations between those map features and the traffic violations, wherein once those map features are identified and weighted in the at least one model, the at least one model is configured to generate the probability score based on the acquired map feature data.
3. The method of claim 2, wherein the map feature data comprises at least two of data related to a functional class of a road, category of a road, a number of lanes, a speed limit, a travel direction, and a presence of a physical divider.
4. The method of claim 1, wherein the traffic violation hotspot comprises a respective location that exhibits a higher than average number of traffic violations than other locations in a region encompassing the respective location.
5. The method of claim 4, wherein the traffic violations include at least one of traffic light jumping, wrong way driving, over speeding, not wearing seatbelt, avoiding stop signs and red lights, distracted driving, passing other vehicles in a no-passing zone.
6. The method of claim 1, wherein the data related to the location comprises sensor data acquired by a vehicle, wherein the at least one model is trained to input sensor data and output the probability score that the location is traffic violation hotspot.
7. The method of claim 6, wherein the sensor data comprises image data acquired by the vehicle.
8. The method of claim 1, wherein the first probability score and the second probability score are weighted equally.
9. The method of claim 1, further comprising: displaying a map comprising a plurality of probabilities of an accident happening at a plurality of locations including at least the location.
10. An apparatus comprising: at least one processor; and at least one memory including computer program code for one or more programs; the at least one memory configured to store the computer program code configured to, with the at least one processor, cause the at least one processor to: acquire data related to a location of a roadway, wherein the data related to the location comprises map feature data and sensor data acquired by a vehicle; generate, using at least one model trained using machine learning, a probability score on how probable the location is to be a traffic violation hotspot based on the acquired data, wherein the at least one model comprises a first model and a second model, wherein the first model is trained to learn which map features are the most relevant

based on correlations between those map features and traffic violations, wherein once those map features are identified and weighted in the at least one model, the at least one model is configured to generate a first probability score based on the acquired map feature data, wherein the second model is trained to input sensor data and output a second probability score that the location is traffic violation hotspot, wherein the probability score is calculated as a weighted average of the first probability score and the second probability score; determine, that the probability score exceeds a threshold score; acquire feature data about the location and a region encompassing the location; generate, by a machine learning model, a probability of an accident occurring at the location; and provide one or more instructions for a vehicle to perform a maneuver for the location based on the probability.

11. The apparatus of claim 10, wherein the data related to the location comprises map feature data, wherein the at least one model is trained to learn which map features are the most relevant based on the correlations between those map features and the traffic violations, wherein once those map features are identified and weighted in the at least one model, the at least one model is configured to generate the probability score based on the acquired map feature data.

12. The apparatus of claim 11, wherein the map feature data comprises at least two of data related to a functional class of a road, category of a road, a number of lanes, a speed limit, a travel direction, and a presence of a physical divider.

13. The apparatus of claim 10, wherein the traffic violation hotspot comprises a respective location that exhibits a higher than average number of traffic violations than other locations in a region encompassing the respective location.

14. The apparatus of claim 13, wherein the traffic violations include at least one of traffic light jumping, wrong way driving, over speeding, not wearing seatbelt, avoiding stop signs and red lights, distracted driving, passing other vehicles in a no-passing zone.

15. The apparatus of claim 10, wherein the data related to the location comprises sensor data acquired by a vehicle, wherein the at least one model is trained to input sensor data and output the probability score that the location is traffic violation hotspot.

16. The apparatus of claim 15, wherein the sensor data comprises image data acquired by the vehicle.

17. The apparatus of claim 10, wherein the first probability score and the second probability score are weighted equally.

18. The apparatus of claim 10, further comprising: displaying a map comprising a plurality of probabilities of an accident happening at a plurality of locations including at least the location.

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